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From Digital and Green Capabilities to Sustainable Value: Green Innovation and Circular Economy Pathways in Indonesian Creative MSMEs

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Corresponding Author:	Isfenti Sadalia Faculty of Vocational, Universitas Sumatera Utara
First Author:	Isfenti Sadalia
Order of Authors:	Isfenti Sadalia Abd Rahman Kadir Imas Purnamasari Jumadil Saputra Wina Nurfitriani Nindy Erisma
Abstract:	Creative micro, small, and medium enterprises (MSMEs) face simultaneous pressure to digitalize and reduce environmental impacts, yet the mechanisms through which digital and green capabilities translate into sustainable business outcomes remain insufficiently understood. Drawing on dynamic capabilities and the natural-resource-based view, this study tests a capability-conversion framework using survey data from 300 creative MSMEs in Bandung, Makassar, and Medan, Indonesia, analyzed with partial least squares structural equation modeling. Sustainable digital maturity, strategic digital leadership, and green entrepreneurial orientation positively predict both green innovation and circular economy practices. Sustainable digital maturity also has a positive direct association with business sustainability, whereas the direct effects of strategic digital leadership and green entrepreneurial orientation are not significant. Green innovation is the strongest direct predictor of business sustainability and mediates the effects of all three antecedent capabilities. Circular economy practices mediate the effects of strategic digital leadership and green entrepreneurial orientation, but not sustainable digital maturity. Government support and regulation neither has a significant direct effect nor strengthens all capability pathways; it significantly strengthens only the strategic digital leadership–business sustainability relationship. The findings advance innovation knowledge by showing that organizational capabilities create sustainable value primarily when converted into concrete green innovations and circular practices, while public support operates as a selective rather than universal boundary condition.
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Dear Editor,

We are pleased to submit our manuscript entitled “**From Digital and Green Capabilities to Sustainable Value: Green Innovation and Circular Economy Pathways in Indonesian Creative MSMEs**” for consideration as a **Regular Article** in the *Journal of Innovation & Knowledge*.

This study examines how sustainable digital maturity, strategic digital leadership, and green entrepreneurial orientation are translated into business sustainability through green innovation and circular economy practices in creative micro, small, and medium-sized enterprises (MSMEs). Using partial least squares structural equation modeling (PLS-SEM) on data from 300 creative MSMEs across Bandung, Makassar, and Medan, Indonesia, the findings show that green innovation represents the strongest direct pathway to business sustainability and significantly mediates the effects of all three antecedent capabilities. Circular economy practices additionally mediate the effects of strategic digital leadership and green entrepreneurial orientation. The results also reveal that government support and regulation operates selectively, significantly strengthening the relationship between strategic digital leadership and business sustainability rather than uniformly enhancing all capability pathways.

The manuscript makes a mechanism-based contribution by distinguishing green innovation and circular economy practices as parallel yet non-equivalent pathways through which digital and green capabilities are translated into sustainable value. It further identifies government support as a selective institutional boundary condition rather than a universally effective external driver. These findings contribute to understanding why the possession of digital capabilities or environmentally oriented strategic postures does not automatically generate sustainable outcomes unless those capabilities are converted into substantive innovation and circular business practices.

We believe that the manuscript is well aligned with the scope of the *Journal of Innovation & Knowledge*, particularly its interests in innovation, SMEs, digital transformation, green innovation, dynamic capabilities, entrepreneurial orientation, and sustainable development. Beyond its Indonesian setting, the study provides insights relevant to resource-constrained firms and emerging economies more broadly by explaining how organizational capabilities can be transformed into sustainable value through innovation-based mechanisms.

We confirm that this manuscript is original, has not been published previously in a form prohibited by Elsevier’s publication policies, and is not currently under consideration for publication elsewhere. All authors have read and approved the submitted version of the manuscript, the authorship order, and its submission to the *Journal of Innovation & Knowledge*.

This research was supported by the **Riset Kolaborasi Indonesia (RKI) program**. Information regarding competing interests and individual author contributions is provided separately in accordance with the journal’s submission requirements.

Thank you for considering our manuscript for publication in the *Journal of Innovation & Knowledge*. We appreciate the opportunity for editorial assessment and peer review.

Sincerely,

Isfenti Sadalia

Corresponding Author

Faculty of Vocational

Universitas Sumatera Utara

Indonesia

Email: isfenti@usu.ac.id

From Digital and Green Capabilities to Sustainable Value: Green Innovation and Circular Economy Pathways in Indonesian Creative MSMEs

Isfenti Sadalia^{1,*}, Abd Rahman Kadir², Imas Purnamasari³, Jumadil Saputra⁴, Wina Nurfitriani¹, Nindy Erisma¹

Affiliations

¹ Faculty of Vocational, Universitas Sumatera Utara, Jl. Dr. T. Mansur No. 9, Kampus Padang Bulan, Medan 20155, Sumatera Utara, Indonesia

² Faculty of Economics and Business, Universitas Hasanuddin, Jl. Perintis Kemerdekaan, Tamalanrea Indah, Tamalanrea, Makassar 90245, South Sulawesi, Indonesia

³ Faculty of Economics and Business Education, Universitas Pendidikan Indonesia, Jl. Dr. Setiabudhi No. 229, Bandung 40154, West Java, Indonesia

⁴ Faculty of Economics and Business, Universiti Malaysia Sarawak, 94300 Kota Samarahan, Sarawak, Malaysia

Corresponding Author

* **Isfenti Sadalia**, Faculty of Vocational, Universitas Sumatera Utara, Jl. Dr. T. Mansur No. 9, Kampus Padang Bulan, Medan 20155, Sumatera Utara, Indonesia. Email: isfenti@usu.ac.id

Author Email Addresses

Isfenti Sadalia: isfenti@usu.ac.id; Abd Rahman Kadir: rahmankadir@feunhas.ac.id; Imas Purnamasari: imaspurnamasari@upi.edu

Jumadil Saputra: jumadilsaputra85@gmail.com; Wina Nurfitriani: winanurfitriani@usu.ac.id; Nindy Erisma: ninerisma@usu.ac.id

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Declaration of Competing Interests

A separate Declaration of Competing Interests file is provided with the submission.

CRedit Authorship Contribution Statement

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Abstract

Creative micro, small, and medium enterprises (MSMEs) face simultaneous pressure to digitalize and reduce environmental impacts, yet the mechanisms through which digital and green capabilities translate into sustainable business outcomes remain insufficiently understood. Drawing on dynamic capabilities and the natural-resource-based view, this study tests a capability-conversion framework using survey data from 300 creative MSMEs in Bandung, Makassar, and Medan, Indonesia, analyzed with partial least squares structural equation modeling. Sustainable digital maturity, strategic digital leadership, and green entrepreneurial orientation positively predict both green innovation and circular economy practices. Sustainable digital maturity also has a positive direct association with business sustainability, whereas the direct effects of strategic digital leadership and green entrepreneurial orientation are not significant. Green innovation is the strongest direct predictor of business sustainability and mediates the effects of all three antecedent capabilities. Circular economy practices mediate the effects of strategic digital leadership and green entrepreneurial orientation, but not sustainable digital maturity. Government support and regulation neither has a significant direct effect nor strengthens all capability pathways; it significantly strengthens only the strategic digital leadership–business sustainability relationship. The findings advance innovation knowledge by showing that organizational capabilities create sustainable value primarily when converted into concrete green innovations and circular practices, while public support operates as a selective rather than universal boundary condition.

Keywords: Digital transformation; Green innovation; Circular economy; Dynamic capabilities; Business sustainability; Creative MSMEs

JEL Classification: O32; Q56; L26

1. Introduction

Digital transformation and environmental sustainability are increasingly intertwined in the innovation strategies of small firms. For resource-constrained MSMEs, however, adopting digital technologies is not equivalent to creating sustainable value. Digital infrastructures, leadership capabilities, and strategic orientations must be translated into changes in products, processes, resource use, and business models before they can materially affect sustainability. Recent research in the *Journal of Innovation & Knowledge* similarly emphasizes that enterprise digitalization can stimulate green innovation and that firm-level digital capabilities become more consequential when they support learning and collaborative innovation processes (Ning et al., 2023; Xie & Wang, 2025).

This capability-conversion problem is particularly relevant to creative MSMEs. Creative firms depend on knowledge-intensive activities, rapid product renewal, market responsiveness, and digital interaction, while many also face growing expectations to reduce waste, extend product life cycles, and improve environmental performance. A systematic review of SMEs' digital transformation has highlighted the importance of organizational knowledge processes and capability development, rather than technology adoption alone, in explaining transformation outcomes (Hafeez et al., 2025). At the same time, research on green innovation shows that environmental innovation can become a source of competitive and sustainability benefits when firms embed it in operational routines and strategic choices (Chen et al., 2006; Hart & Dowell, 2011).

The present study focuses on three organizational antecedents that capture complementary aspects of this transformation: sustainable digital maturity (SDM), strategic digital leadership (SDL), and green entrepreneurial orientation (GEO). SDM reflects the extent to which digital systems and infrastructure are sufficiently developed to support organizational and sustainability objectives; SDL captures leadership direction and capability deployment in digital transformation; and GEO represents a proactive, innovative, and risk-oriented environmental posture. Existing studies have examined related links separately, but the combined mechanism through which these capabilities are converted into green innovation (GI), circular economy practices (CEP), and ultimately business sustainability (BS) remains less clear, particularly in creative-industry MSMEs.

Two transmission mechanisms are central to the proposed framework. Green innovation represents tangible changes to products and processes intended to reduce environmental impacts, whereas circular economy practices emphasize resource efficiency, reuse, recycling, and value retention. The distinction matters because the two mechanisms may not transmit digital and green capabilities in the same way. Evidence that digitalization supports green innovation (Ning et al., 2023), that green relationship learning links digital capabilities with collaborative green innovation (Xie & Wang, 2025), and that circular practices have distinct organizational antecedents (Tritto et al., 2024) suggests that a dual-mechanism model can reveal how sustainable value is actually produced rather than merely whether capabilities correlate with sustainability.

The study also examines government support and regulation (GSR). Public policy can reduce uncertainty and facilitate eco-innovation adoption, yet support need not strengthen every firm-level capability uniformly. Evidence from Indonesian SMEs shows that government support and eco-regulation can operate through intervening environmental mechanisms rather than as simple direct performance drivers (Achmad et al., 2023). This motivates a boundary-condition perspective in which GSR may complement some internal capabilities more strongly than others.

Empirically, the study uses a pooled sample of 300 creative MSMEs—100 each from Bandung, Makassar, and Medan. These cities provide strategically diverse urban settings across Java, Sulawesi, and Sumatra without implying national statistical representativeness. The study contributes to innovation and knowledge research in four ways. First, it frames sustainability as a capability-conversion problem rather than a direct consequence of digitalization. Second, it distinguishes green innovation and circular economy practices as parallel but non-identical transmission mechanisms. Third, it evaluates GSR as a selective institutional boundary condition. Fourth, it uses multi-city evidence from an emerging economy to identify lessons about how resource-constrained creative firms can convert digital and environmental capabilities into sustainable value. Accordingly, the study asks how SDM, SDL, and GEO shape GI and CEP; how these mechanisms transmit effects to BS; and whether GSR directly or conditionally strengthens sustainable outcomes.

2. Literature Review and Hypotheses Development

2.1. Theoretical framework: dynamic capabilities, the natural-resource-based view, and institutional conditions

Dynamic capabilities theory explains how firms sense changes, seize opportunities, and reconfigure resources in volatile environments (Teece, 2007). In the present framework, SDM and SDL represent complementary digital capabilities. SDM captures the organizational readiness and infrastructure that make digital reconfiguration possible, whereas SDL captures the strategic direction through which digital resources are mobilized. For MSMEs, these capabilities are expected to matter not because digitalization is inherently sustainable, but because it can enable new combinations of information, processes, and organizational routines that support environmental innovation.

The natural-resource-based view (NRBV) extends resource-based reasoning by treating the natural environment as a central constraint and opportunity for competitive advantage (Hart, 1995; Hart & Dowell, 2011). GEO reflects a strategic posture that directs attention toward environmental opportunities, while GI and CEP represent operational manifestations of that posture. By combining dynamic-capability and NRBV reasoning, the framework proposes a conversion sequence: digital and green capabilities create potential, GI and CEP operationalize that potential, and BS captures the resulting economic, social, and environmental value.

Institutional conditions are incorporated as a boundary rather than as a substitute for internal capability. Government incentives, regulation, technical support, and capacity-building can reduce uncertainty and facilitate eco-innovation, but their effectiveness depends on how firms use them (Achmad et al., 2023). This logic allows GSR to have a direct relationship with BS while also potentially strengthening the ability of SDM, SDL, and GEO to generate sustainable outcomes. Figure 1 summarizes the proposed framework.

Figure 1. Conceptual framework and hypotheses.

2.2. Sustainable digital maturity

Sustainable digital maturity refers to the degree to which digital technologies, data, organizational routines, and supporting infrastructure are integrated in ways that can also serve sustainability objectives. Maturity is therefore more than the possession of digital tools; it reflects an organization's capacity to align technological development with strategic priorities. A sustainable digital-transformation roadmap for SMEs similarly emphasizes the alignment of digital maturity with strategy and sustainability (Mick et al., 2024).

Higher SDM should make environmental information more visible, improve process traceability, and expand the firm's capacity to redesign products and workflows. These capabilities can support GI by facilitating experimentation and data-enabled process improvement, and they can support CEP by improving resource tracking and coordination. Consistent with evidence that enterprise digitalization can improve green innovation (Ning et al., 2023), SDM is also expected to contribute to BS when digital systems are sufficiently integrated into value-creating activities.

H1: Sustainable digital maturity has a positive and significant effect on green innovation.

H2: Sustainable digital maturity has a positive and significant effect on circular economy practices.

H3: Sustainable digital maturity has a positive and significant effect on business sustainability.

2.3. Strategic digital leadership

Strategic digital leadership concerns the ability of leaders to articulate a digital vision, allocate resources, develop capabilities, and coordinate organizational change. From a dynamic-capability perspective, leadership helps convert technological possibilities into purposeful reconfiguration. Empirical research has linked digital leadership with green innovation and firm sustainability, indicating that leadership can make digital transformation more consequential when it is connected to environmental innovation (Zada et al., 2025).

In creative MSMEs, SDL can direct scarce resources toward eco-design, digital monitoring, cleaner processes, and circular business practices. It can also coordinate learning across employees, suppliers, and market partners. Digital capabilities have been shown to support green collaborative innovation when they are combined with relationship learning (Xie & Wang, 2025), which reinforces the expectation that leadership should influence GI and CEP. A direct relationship with BS is also proposed because coherent digital leadership may improve strategic agility and resource deployment even before specific environmental outcomes are realized.

H4: Strategic digital leadership has a positive and significant effect on green innovation.

H5: Strategic digital leadership has a positive and significant effect on circular economy practices.

H6: Strategic digital leadership has a positive and significant effect on business sustainability.

2.4. Green entrepreneurial orientation

Green entrepreneurial orientation extends entrepreneurial orientation by embedding environmental considerations in innovativeness, proactiveness, and risk taking. It represents a willingness to identify ecological challenges as entrepreneurial opportunities and to commit resources before outcomes are fully certain. Reviews of green entrepreneurial orientation emphasize its relevance for connecting entrepreneurial behavior with environmental innovation (Golsefid-Alavi et al., 2021).

A stronger GEO should increase the likelihood that firms develop greener products and processes and adopt circular practices that retain material value. Research in emerging economies also links GEO with green innovation capability (Rong et al., 2025). Within the NRBV, this posture can support sustainable advantage when it is translated into resource-saving and pollution-preventing routines. The framework therefore proposes positive relationships from GEO to GI, CEP, and BS.

H7: Green entrepreneurial orientation has a positive and significant effect on green innovation.

H8: Green entrepreneurial orientation has a positive and significant effect on circular economy practices.

H9: Green entrepreneurial orientation has a positive and significant effect on business sustainability.

2.5. Green innovation and circular economy practices

Green innovation comprises product and process innovations that reduce environmental harm while preserving or enhancing organizational value. Its strategic relevance follows from the possibility that environmentally oriented innovation can lower resource costs, strengthen differentiation, and improve stakeholder legitimacy (Chen et al., 2006). In the present model, GI is expected to connect upstream capabilities with multidimensional BS because it translates technological and strategic potential into observable changes in products and processes.

Circular economy practices emphasize resource efficiency and value retention through reduction, reuse, recycling, recovery, and related operational arrangements. Survey evidence on SMEs shows that circular practices vary systematically across organizational contexts and are closely connected to broader sustainability goals (Tritto et al., 2024). Because circularity can reduce material losses and encourage more resilient operating models, CEP is expected to contribute positively to BS.

H10: Green innovation has a positive and significant effect on business sustainability.

H11: Circular economy practices have a positive and significant effect on business sustainability.

2.6. Direct role of government support and regulation

Government support and regulation can shape the incentives, information, and resources available for green transformation. Financial assistance, technical programs, regulatory clarity, and environmental standards may lower adoption barriers and make sustainability investments more feasible for resource-constrained firms. Evidence from Indonesian SMEs indicates that government support and eco-regulation are relevant to eco-innovation adoption, although their effects may operate through organizational and environmental mechanisms (Achmad et al., 2023). A positive direct association with BS is therefore proposed.

H12: Government support and regulation have a positive and significant direct effect on business sustainability.

2.7. Mediating roles of green innovation and circular economy practices

The capability-conversion perspective implies that SDM, SDL, and GEO may influence BS indirectly by enabling concrete environmental actions. GI is one such mechanism because digital systems, leadership, and green strategic posture can improve the firm's capacity to redesign products and processes. This reasoning is

consistent with research showing that green innovation can transmit the sustainability effects of digital leadership (Zada et al., 2025).

CEP provides a second, distinct mechanism. Digital systems can support resource visibility, leaders can institutionalize circular routines, and GEO can motivate proactive resource recirculation. Because GI and CEP emphasize different forms of operationalization, they are tested as parallel mediators rather than treated as interchangeable manifestations of sustainability.

H13: Green innovation mediates the relationship between sustainable digital maturity and business sustainability.

H14: Circular economy practices mediate the relationship between sustainable digital maturity and business sustainability.

H15: Green innovation mediates the relationship between strategic digital leadership and business sustainability.

H16: Circular economy practices mediate the relationship between strategic digital leadership and business sustainability.

H17: Green innovation mediates the relationship between green entrepreneurial orientation and business sustainability.

H18: Circular economy practices mediate the relationship between green entrepreneurial orientation and business sustainability.

2.8. Moderating role of government support and regulation

If external support complements internal capabilities, GSR should strengthen the conversion of SDM, SDL, and GEO into BS. For example, policy incentives may improve the returns to digitally enabled sustainability investments, regulatory clarity may help leaders prioritize transformation, and government programs may reduce the uncertainty faced by environmentally proactive entrepreneurs. At the same time, such complementarity need not be uniform, which makes the three moderation hypotheses empirically informative rather than assuming a universal policy effect.

H19: Government support and regulation positively moderate the relationship between sustainable digital maturity and business sustainability.

H20: Government support and regulation positively moderate the relationship between strategic digital leadership and business sustainability.

H21: Government support and regulation positively moderate the relationship between green entrepreneurial orientation and business sustainability.

3. Methodology

3.1. Research design and sample

The study used a cross-sectional survey design among creative-industry MSMEs in Bandung, Makassar, and Medan, Indonesia. The pooled sample comprised 300 MSMEs, with 100 observations from each city. The three urban hubs were selected to provide geographically and economically diverse settings across Java, Sulawesi, and Sumatra. The pooled design is used to estimate a common structural model across these urban creative-industry settings; it is not presented as a probability-based national sample.

The study employed purposive sampling to recruit owners and managers of creative-industry MSMEs in Bandung, Makassar, and Medan. Respondents were selected based on their involvement in the management and strategic decision-making of their businesses. Data collection commenced in January 2025 and was conducted online using a structured questionnaire distributed through Google Forms. A total of 300 usable responses were included in the analysis, comprising 100 respondents from each city. Participation was

voluntary, and respondents were informed about the purpose of the study and the confidentiality of their responses before completing the questionnaire. Informed consent was obtained from all participants prior to participation.

3.2. Measures

All constructs were measured using a five-point Likert-type response format. The study adapted established scales to the creative-MSME context. Sustainable Digital Maturity was measured with 12 items adapted from Mick et al. (2024); Strategic Digital Leadership with nine items adapted from Zada et al. (2025); Green Entrepreneurial Orientation with eight items adapted from Golsefid-Alavi et al. (2021), as operationalized in later work including Rong et al. (2025); Green Innovation with six items adapted from Chen et al. (2006); Circular Economy Practices with seven items adapted from Tritto et al. (2024); Government Support and Regulation with six items adapted from Achmad et al. (2023); and Business Sustainability with nine items drawing on the triple-bottom-line perspective and SME sustainability measures (Baeshen et al., 2021; Elkington, 1997; Fandeli et al., 2026).

The questionnaire was developed and administered in Bahasa Indonesia, with the measurement items adapted and contextually adjusted to ensure their relevance to creative-industry MSMEs in Indonesia while preserving the conceptual meaning of the original constructs. Prior to the main survey, the questionnaire was pilot-tested with 30 respondents to assess the clarity, comprehensibility, and suitability of the items for the target respondents. The pilot test provided an initial assessment of the questionnaire before its use in the main data collection involving 300 MSMEs.

3.3. Data analysis

Descriptive information and distributional checks were conducted using IBM SPSS Statistics version 26, while the measurement and structural models were evaluated using SmartPLS 4. Partial least squares structural equation modeling (PLS-SEM) was selected because the proposed model contains multiple latent constructs, parallel mediating mechanisms, and interaction terms and because PLS-SEM is well suited to prediction-oriented and complex structural models (Hair et al., 2019, 2022).

The measurement model was assessed using indicator loadings, average variance extracted (AVE), composite reliability (CR), Cronbach's alpha (CA), and the heterotrait-monotrait ratio (HTMT) of correlations (Fornell & Larcker, 1981; Henseler et al., 2015). Structural collinearity was assessed using inner variance inflation factor (VIF) values. Because the available output reports inner VIF rather than a separate full-collinearity assessment, these VIF statistics are interpreted as structural collinearity diagnostics and are not used as standalone evidence that common method bias is absent.

The structural model was evaluated using path coefficients, t-statistics, p-values, coefficients of determination (R^2), Stone-Geisser Q^2 values from blindfolding (omission distance = 7), and standardized root mean square residual (SRMR). Mediation was assessed using bootstrapped specific indirect effects, and moderation was evaluated using interaction terms in SmartPLS. Bootstrapping with 5,000 subsamples was used to assess the statistical significance of the direct, indirect, and moderating effects. Statistical significance was evaluated at $p < 0.05$.

4. Results

4.1. Measurement model

Table 1 reports indicator reliability, convergent validity, and internal consistency. Most outer loadings exceed 0.70. Seven indicators fall slightly below 0.70—SDL3 (0.654), SDL6 (0.675), GEO1 (0.691), GEO2 (0.699), GEO7 (0.659), GSR1 (0.667), and GSR3 (0.690). These indicators were retained because construct-level convergent validity and reliability remain satisfactory: all AVE values exceed 0.50, CR ranges from

0.874 to 0.984, and CA ranges from 0.829 to 0.982. The results therefore support adequate convergent validity and internal consistency at the construct level.

Discriminant validity was evaluated using HTMT (Table 2). All HTMT values are below 0.85, and the highest observed value is 0.558 between GI and BS. The constructs are therefore empirically distinguishable under the applied criterion (Henseler et al., 2015).

4.2. Structural model diagnostics

The inner VIF values in Table 3 range from 1.190 to 1.526, indicating no problematic structural collinearity among the predictors. These values are used only for multicollinearity assessment within the structural model; they are not interpreted as a complete common-method-bias test.

Table 4 shows that the model explains 42.1% of the variance in BS (adjusted $R^2 = 0.403$), 25.6% of the variance in CEP (adjusted $R^2 = 0.249$), and 19.9% of the variance in GI (adjusted $R^2 = 0.191$). The corresponding Q^2 values in Table 5 are positive for all endogenous constructs—0.342 for BS, 0.145 for CEP, and 0.114 for GI—indicating predictive relevance under the blindfolding criterion, with the strongest value observed for BS. The results are reported without overstating these values as evidence of out-of-sample prediction, which would require a dedicated predictive procedure.

The saturated-model SRMR is 0.040 and the estimated-model SRMR is 0.049 (Table 6), both below the commonly used approximate-fit benchmark of 0.08. However, the estimated-model SRMR (0.049) exceeds the reported bootstrap 95% HI (0.044) and 99% HI (0.046). Accordingly, the conventional SRMR benchmark suggests acceptable approximate fit, whereas the bootstrap HI comparison is more conservative for the estimated model. SRMR is therefore treated as supplementary rather than conclusive evidence of model adequacy.

4.3. Hypothesis testing, mediation, and moderation

The bootstrapping results are summarized in Table 7. SDM has positive associations with GI ($\beta = 0.156$, $t = 2.757$, $p = 0.006$), CEP ($\beta = 0.124$, $t = 2.114$, $p = 0.035$), and BS ($\beta = 0.145$, $t = 2.014$, $p = 0.044$), supporting H1–H3. SDL is positively associated with GI ($\beta = 0.172$, $t = 3.014$, $p = 0.003$) and CEP ($\beta = 0.231$, $t = 4.474$, $p < 0.001$), supporting H4 and H5, but its direct association with BS is not significant ($\beta = -0.031$, $t = 0.520$, $p = 0.603$), so H6 is not supported. GEO is positively associated with GI ($\beta = 0.258$, $t = 5.073$, $p < 0.001$) and CEP ($\beta = 0.301$, $t = 5.571$, $p < 0.001$), supporting H7 and H8, while its direct relationship with BS is not significant ($\beta = 0.064$, $t = 1.315$, $p = 0.189$), so H9 is not supported.

GI has the largest direct coefficient on BS ($\beta = 0.384$, $t = 7.654$, $p < 0.001$), supporting H10. CEP also has a positive relationship with BS ($\beta = 0.202$, $t = 3.526$, $p < 0.001$), supporting H11. GSR has no significant direct association with BS ($\beta = 0.036$, $t = 0.780$, $p = 0.435$), so H12 is not supported.

For mediation, the $SDM \rightarrow GI \rightarrow BS$ indirect effect is significant ($\beta = 0.060$, $t = 2.647$, $p = 0.008$), supporting H13. Because SDM also retains a significant direct effect on BS, this pattern is consistent with complementary/partial mediation. The $SDM \rightarrow CEP \rightarrow BS$ indirect effect is not significant at the 5% level ($\beta = 0.025$, $t = 1.750$, $p = 0.080$), so H14 is not supported. Both SDL indirect effects are significant through GI ($\beta = 0.066$, $t = 2.713$, $p = 0.007$) and CEP ($\beta = 0.047$, $t = 2.782$, $p = 0.005$), supporting H15 and H16. Because the direct $SDL \rightarrow BS$ relationship is not significant, these results are consistent with indirect-only mediation. The same pattern holds for GEO: the indirect effects through GI ($\beta = 0.099$, $t = 4.023$, $p < 0.001$) and CEP ($\beta = 0.061$, $t = 2.889$, $p = 0.004$) are significant while the direct $GEO \rightarrow BS$ effect is not, supporting H17 and H18 and indicating indirect-only mediation.

For moderation, GSR does not significantly moderate $SDM \rightarrow BS$ ($\beta = 0.079$, $t = 1.442$, $p = 0.150$) or $GEO \rightarrow BS$ ($\beta = 0.079$, $t = 1.337$, $p = 0.181$), so H19 and H21 are not supported. In contrast, GSR positively moderates $SDL \rightarrow BS$ ($\beta = 0.154$, $t = 2.511$, $p = 0.012$), supporting H20. This selective moderation pattern is central to the interpretation developed below.

5. Discussion

5.1. Converting digital and green capabilities into environmental innovation

The direct paths to GI and CEP show a consistent pattern: SDM, SDL, and GEO all strengthen both environmental mechanisms. This is important because it shifts the interpretation of digital transformation away from technology possession and toward capability deployment. The SDM findings are consistent with the idea that digital maturity creates an infrastructure for environmental experimentation and resource visibility (Mick et al., 2024), while the SDL findings indicate that leadership is needed to direct such infrastructure toward strategic environmental uses (Zada et al., 2025). GEO has the largest coefficients among the three antecedents for both GI and CEP, suggesting that an explicitly environmental entrepreneurial posture is especially important in converting opportunities into green and circular actions.

These results also connect with the innovation-and-knowledge literature. Digitalization can facilitate green innovation (Ning et al., 2023), but the pathway is strengthened when digital capability supports learning and collaboration (Xie & Wang, 2025). The present findings extend that logic to creative MSMEs by showing that three different capability domains—digital maturity, digital leadership, and green entrepreneurial orientation—converge on two operational mechanisms. In other words, the relevant knowledge contribution is not simply that digitalization 'matters,' but that the sustainability value of digital and green capabilities depends on how firms convert them into innovation and circular routines.

5.2. Green innovation and circularity as distinct capability-conversion mechanisms

GI is the strongest direct predictor of BS in the model, while CEP also contributes positively. The mediation results reveal that these mechanisms are not interchangeable. GI transmits the effects of SDM, SDL, and GEO, whereas CEP transmits the effects of SDL and GEO but not SDM. This asymmetry suggests that digital maturity alone may be sufficient to support data-enabled product and process innovation, yet the conversion of digital maturity into circular operating routines may require additional leadership, organizational commitment, or ecosystem coordination beyond technological readiness.

The pattern is especially informative for SDL and GEO. Neither construct has a significant direct relationship with BS, but both have significant indirect relationships through GI and CEP. Their sustainability relevance therefore lies in what they cause the firm to do rather than in the capability or orientation by itself. This provides a more specific interpretation of the 'black box' between capabilities and performance. Consistent with the NRBV, environmental value arises when strategic resources are embedded in pollution-preventing, resource-saving, and product-stewardship activities (Hart, 1995; Hart & Dowell, 2011). It also aligns with evidence that digital leadership can enhance sustainability through green innovation (Zada et al., 2025).

SDM differs because it retains a small but significant direct relationship with BS in addition to the GI-mediated path. This complementary/partial mediation implies that mature digital systems may generate some sustainability value through operational efficiency and coordination even when not fully expressed as GI. Yet the stronger GI pathway reinforces the broader conversion argument: the greatest sustainability gains arise when technological capability becomes substantive environmental innovation.

5.3. Government support as a selective institutional boundary condition

GSR has no significant direct relationship with BS and does not strengthen the SDM or GEO pathways. It does, however, strengthen the SDL → BS relationship. This selective result cautions against treating government support as a universal performance lever. Public programs may be most effective when firm leaders possess the strategic capacity to interpret incentives, coordinate organizational change, and connect policy opportunities with business decisions. In that sense, policy complements leadership rather than replacing internal capability.

This interpretation is compatible with evidence that government support and eco-regulation can influence eco-innovation through intervening mechanisms (Achmad et al., 2023). The non-significant moderation results for SDM and GEO are therefore theoretically useful rather than failed hypotheses: they define the limits of the institutional effect. Digital infrastructure and entrepreneurial green orientation may be driven more strongly by internal routines and values, while strategic digital leadership appears more responsive to external policy support.

5.4. Contributions to innovation and knowledge

- **Capability-conversion contribution.** The study shows that digital and green capabilities do not uniformly produce sustainability directly. Their effects depend substantially on conversion into green innovation and circular economy practices, providing a mechanism-based explanation of sustainable value creation.
- **Dual-mechanism contribution.** GI and CEP are distinct but complementary pathways. Their different mediation patterns demonstrate that environmental innovation and circularity should not be collapsed into a single generic sustainability mechanism.
- **Boundary-condition contribution.** GSR operates selectively: it strengthens the SDL → BS relationship but not the SDM or GEO pathways and has no direct effect on BS. This qualifies assumptions that policy support universally amplifies sustainable performance.
- **Context-to-theory contribution.** Multi-city evidence from Indonesian creative MSMEs extends capability and NRBV reasoning to resource-constrained, knowledge-intensive firms in an emerging economy. The wider lesson is that digital transformation creates sustainable value when it becomes organizational knowledge-in-use—embedded in products, processes, circular routines, and leadership decisions rather than remaining as technological potential.

6. Implications

6.1. Managerial implications

Managers should evaluate digital investments by the environmental actions they enable. Because GI is the strongest direct predictor of BS and mediates all three antecedent capabilities, digital systems should be explicitly connected to eco-design, cleaner processes, waste reduction, and other verifiable innovation activities rather than treated as an end in themselves.

Leadership development deserves equal attention to technology acquisition. SDL predicts both GI and CEP, and its sustainability effect becomes significant through these mechanisms. Creative-MSME leaders should therefore build the ability to allocate digital resources toward green projects, coordinate experimentation, and institutionalize circular routines. GEO should likewise be translated into operating practices; environmental proactiveness generates value when it leads to innovation and circularity.

6.2. Policy implications

Policy support should be targeted rather than generic. The only significant moderation effect is $GSR \times SDL \rightarrow BS$, suggesting that public interventions may have greater leverage when they strengthen leaders' ability to convert digital opportunities into sustainable business decisions. Capacity-building programs that integrate digital strategy, environmental innovation, and circular business design are therefore more closely aligned with the empirical results than undifferentiated support.

The absence of broad direct and moderating effects also implies that policymakers should not equate program availability with firm-level sustainability. Implementation should be designed around uptake, managerial usability, and the conversion of support into specific innovation and circular-economy activities.

7. Limitations and Future Research

The study has several limitations. First, the cross-sectional design supports association-based structural inference but does not establish temporal causality. Longitudinal research could examine how capability conversion develops as firms mature digitally and environmentally. Second, the measures are self-reported, so common method and perceptual biases remain possible; the inner VIF results should not be interpreted as a complete common-method-bias assessment. Future studies could combine survey data with objective financial, waste, energy, or innovation indicators.

Third, the sample is intentionally multi-city but is limited to Bandung, Makassar, and Medan. These settings provide regional diversity but should not be treated as statistically representative of all Indonesian MSMEs. Replication across rural areas, additional creative clusters, and other ASEAN economies would strengthen external validity. Fourth, future models could examine additional knowledge and ecosystem mechanisms—such as organizational learning, green knowledge sharing, supply-chain integration, or customer environmental pressure—without assuming that these unmeasured constructs explain the present results.

8. Conclusion

This study reframes sustainable digital transformation in creative MSMEs as a capability-conversion process. Across 300 MSMEs in three Indonesian urban creative hubs, SDM, SDL, and GEO are associated with stronger GI and CEP, but their relationships with BS differ markedly. GI is the central transmission mechanism and the strongest direct predictor of BS; CEP provides a second pathway, particularly for SDL and GEO. The non-significant direct effects of SDL and GEO, together with their significant indirect effects, indicate that leadership and environmental orientation create sustainable value when they are operationalized. GSR is similarly selective: it strengthens the SDL pathway but does not directly improve BS or broadly amplify SDM and GEO. The broader contribution to innovation knowledge is therefore a conditional one: digital and green capabilities become valuable for sustainability when firms convert them into concrete innovations and circular routines, and institutional support is most useful when it complements the internal capability needed to act on it.

Data Availability

The anonymized data supporting the findings of this study are available from the corresponding author upon reasonable request.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used OpenAI ChatGPT to support language refinement, structural editing, consistency checking, and alignment with journal requirements. The tool was not used to generate or alter the research data or statistical results. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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Tables

Table 1. Measurement model

Construct	Measurement Items	Loading Factor	AVE	CR	CA
Sustainable Digital Maturity	SDM1	0.885	0.834	0.984	0.982
Sustainable Digital Maturity	SDM2	0.940	0.834	0.984	0.982
Sustainable Digital Maturity	SDM3	0.878	0.834	0.984	0.982
Sustainable Digital Maturity	SDM4	0.922	0.834	0.984	0.982
Sustainable Digital Maturity	SDM5	0.935	0.834	0.984	0.982
Sustainable Digital Maturity	SDM6	0.903	0.834	0.984	0.982
Sustainable Digital Maturity	SDM7	0.912	0.834	0.984	0.982
Sustainable Digital Maturity	SDM8	0.897	0.834	0.984	0.982
Sustainable Digital Maturity	SDM9	0.934	0.834	0.984	0.982
Sustainable Digital Maturity	SDM10	0.928	0.834	0.984	0.982
Sustainable Digital Maturity	SDM11	0.870	0.834	0.984	0.982
Sustainable Digital Maturity	SDM12	0.951	0.834	0.984	0.982
Strategic Digital Leadership	SDL1	0.839	0.568	0.922	0.904
Strategic Digital Leadership	SDL2	0.813	0.568	0.922	0.904
Strategic Digital Leadership	SDL3	0.654	0.568	0.922	0.904
Strategic Digital Leadership	SDL4	0.789	0.568	0.922	0.904
Strategic Digital Leadership	SDL5	0.703	0.568	0.922	0.904
Strategic Digital Leadership	SDL6	0.675	0.568	0.922	0.904
Strategic Digital Leadership	SDL7	0.739	0.568	0.922	0.904
Strategic Digital Leadership	SDL8	0.728	0.568	0.922	0.904
Strategic Digital Leadership	SDL9	0.821	0.568	0.922	0.904
Green Entrepreneurial Orientation	GEO1	0.691	0.579	0.916	0.895
Green Entrepreneurial Orientation	GEO2	0.699	0.579	0.916	0.895
Green Entrepreneurial Orientation	GEO3	0.831	0.579	0.916	0.895
Green Entrepreneurial Orientation	GEO4	0.781	0.579	0.916	0.895
Green Entrepreneurial Orientation	GEO5	0.791	0.579	0.916	0.895
Green Entrepreneurial Orientation	GEO6	0.865	0.579	0.916	0.895
Green Entrepreneurial Orientation	GEO7	0.659	0.579	0.916	0.895
Green Entrepreneurial Orientation	GEO8	0.745	0.579	0.916	0.895
Green Innovation	GI1	0.803	0.589	0.895	0.860
Green Innovation	GI2	0.827	0.589	0.895	0.860
Green Innovation	GI3	0.785	0.589	0.895	0.860
Green Innovation	GI4	0.704	0.589	0.895	0.860
Green Innovation	GI5	0.757	0.589	0.895	0.860
Green Innovation	GI6	0.721	0.589	0.895	0.860
Circular Economy Practices	CEP1	0.729	0.585	0.908	0.881
Circular Economy Practices	CEP2	0.771	0.585	0.908	0.881
Circular Economy Practices	CEP3	0.792	0.585	0.908	0.881
Circular Economy Practices	CEP4	0.789	0.585	0.908	0.881
Circular Economy Practices	CEP5	0.739	0.585	0.908	0.881

Construct	Measurement Items	Loading Factor	AVE	CR	CA
Circular Economy Practices	CEP6	0.822	0.585	0.908	0.881
Circular Economy Practices	CEP7	0.705	0.585	0.908	0.881
Government Support & Regulation	GSR1	0.667	0.536	0.874	0.829
Government Support & Regulation	GSR2	0.775	0.536	0.874	0.829
Government Support & Regulation	GSR3	0.690	0.536	0.874	0.829
Government Support & Regulation	GSR4	0.714	0.536	0.874	0.829
Government Support & Regulation	GSR5	0.784	0.536	0.874	0.829
Government Support & Regulation	GSR6	0.754	0.536	0.874	0.829
Business Sustainability	BS1	0.929	0.849	0.981	0.978
Business Sustainability	BS2	0.916	0.849	0.981	0.978
Business Sustainability	BS3	0.933	0.849	0.981	0.978
Business Sustainability	BS4	0.905	0.849	0.981	0.978
Business Sustainability	BS5	0.899	0.849	0.981	0.978
Business Sustainability	BS6	0.944	0.849	0.981	0.978
Business Sustainability	BS7	0.933	0.849	0.981	0.978
Business Sustainability	BS8	0.906	0.849	0.981	0.978
Business Sustainability	BS9	0.928	0.849	0.981	0.978

Table 2. Heterotrait-monotrait ratio of correlations (HTMT)

	GEO	BS	CEP	GSR	GI	SDL	SDM
GEO							
BS	0.341						
CEP	0.476	0.484					
GSR	0.352	0.297	0.398				
GI	0.425	0.558	0.479	0.350			
SDL	0.437	0.290	0.437	0.361	0.371		
SDM	0.320	0.372	0.318	0.315	0.320	0.383	

Table 3. Structural collinearity statistics (inner VIF)

Construct	Business Sustainability	Circular Economy Practices	Green Innovation
Green Entrepreneurial Orientation	1.415	1.220	1.220
Business Sustainability			
Circular Economy Practices	1.526		
Government Support & Regulation	1.256		
Green Innovation	1.377		
Strategic Digital Leadership	1.457	1.278	1.278
Sustainable Digital Maturity	1.426	1.190	1.190

Table 4. Coefficient of determination (R²)

Construct	R Square	R Square Adjusted
Business Sustainability	0.421	0.403
Circular Economy Practices	0.256	0.249
Green Innovation	0.199	0.191

Table 5. Predictive relevance (Q²) from blindfolding

Construct	SSO	SSE	Q2 (=1-SSE/SSO)
Green Entrepreneurial Orientation	2400.000	2400.000	
Business Sustainability	2700.000	1776.176	0.342
Circular Economy Practices	2100.000	1795.253	0.145
Government Support & Regulation	1800.000	1800.000	
Green Innovation	1800.000	1594.859	0.114
Strategic Digital Leadership	2700.000	2700.000	
Sustainable Digital Maturity	3600.000	3600.000	

Table 6. Standardized root mean square residual (SRMR)

Model	Original Sample (O)	Sample Mean (M)	95% HI	99% HI
Saturated Model	0.040	0.039	0.042	0.044
Estimated Model	0.049	0.041	0.044	0.046

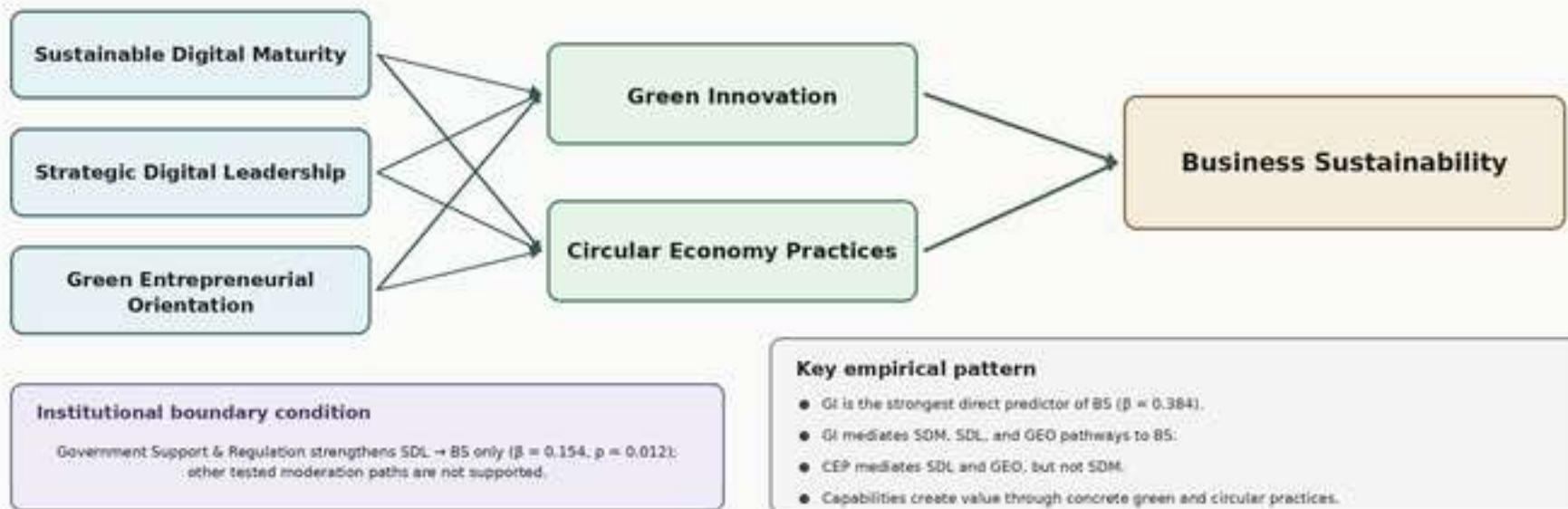
Table 7. Hypothesis testing results

Hyp.	Path	β	T-Stat	p-Value	Decision
Direct Effect					
H1	SDM → GI	0.156	2.757	0.006	Supported
H2	SDM → CEP	0.124	2.114	0.035	Supported
H3	SDM → BS	0.145	2.014	0.044	Supported
H4	SDL → GI	0.172	3.014	0.003	Supported
H5	SDL → CEP	0.231	4.474	<0.001	Supported
H6	SDL → BS	-0.031	0.520	0.603	Not Supported
H7	GEO → GI	0.258	5.073	<0.001	Supported
H8	GEO → CEP	0.301	5.571	<0.001	Supported
H9	GEO → BS	0.064	1.315	0.189	Not Supported
H10	GI → BS	0.384	7.654	<0.001	Supported
H11	CEP → BS	0.202	3.526	<0.001	Supported
H12	GSR → BS	0.036	0.780	0.435	Not Supported
Indirect Effect (Mediation)					
H13	SDM → GI → BS	0.060	2.647	0.008	Supported
H14	SDM → CEP → BS	0.025	1.750	0.080	Not Supported
H15	SDL → GI → BS	0.066	2.713	0.007	Supported
H16	SDL → CEP → BS	0.047	2.782	0.005	Supported

Hyp.	Path	β	T-Stat	p-Value	Decision
H17	$\text{GEO} \rightarrow \text{GI} \rightarrow \text{BS}$	0.099	4.023	<0.001	Supported
H18	$\text{GEO} \rightarrow \text{CEP} \rightarrow \text{BS}$	0.061	2.889	0.004	Supported
Moderating Effect					
H19	$\text{GSR} \times \text{SDM} \rightarrow \text{BS}$	0.079	1.442	0.150	Not Supported
H20	$\text{GSR} \times \text{SDL} \rightarrow \text{BS}$	0.154	2.511	0.012	Supported
H21	$\text{GSR} \times \text{GEO} \rightarrow \text{BS}$	0.079	1.337	0.181	Not Supported

Capability conversion into sustainable value

Evidence from 300 creative MSMEs in Bandung, Makassar, and Medan



Capability-to-innovation-to-sustainability model

